

BESHIR SAAD SALEM

INSTRUMENTATION & CONTROL SPECIALIST · PROCESS ANALYTICAL INSTRUMENTS SUPERVISOR · ACTING HEAD OF INSTRUMENTATION DIVISION

Ras Lanuf City, Libya | +218 91 4860116 | beshir8147@raslanuf.ly | RASCO — Ras Lanuf Oil & Gas Processing Company

PROFESSIONAL PROFILE

Instrumentation and Control engineering professional with nearly 19 years of experience at Ras Lanuf Oil & Gas Processing Company (RASCO), spanning field instrumentation, process analytical instruments, and divisional technical leadership. Proven track record across calibration, commissioning, preventive and shutdown maintenance, technical evaluation, and modernization of critical measurement systems including gas chromatographs, liquid analyzers, control valves, transmitters, and DCS-integrated instrumentation. Acting Head of the Instrumentation Division, leading project planning, vendor communication, procurement support, and reliability initiatives. Combines deep field expertise with engineering studies, technical reporting, and a growing focus on digital transformation and AI-supported reliability tools.

PROFESSIONAL EXPERIENCE

Acting Head of Instrumentation Division

2025 — Present

Ras Lanuf Oil & Gas Processing Company (RASCO), Ras Lanuf, Libya

- ▶ Lead the Instrumentation Division's technical direction, including project planning, procurement support, and vendor negotiation.
- ▶ Sponsor and oversee major modernization projects spanning control valves, radar tank gauging, vibration monitoring, and legacy pneumatic system replacement.
- ▶ Mentor instrumentation technicians and drive digital transformation initiatives across the division.
- ▶ Serve as primary technical point of contact for engineering studies, reliability improvement programs, and cross-functional coordination.

Process Analytical Instruments Specialist / Supervisor

2020 — 2025

Ras Lanuf Oil & Gas Processing Company (RASCO), Ras Lanuf, Libya

- ▶ Held technical ownership of the plant's analytical instrumentation, including gas chromatographs, pH and conductivity analyzers, sample conditioning systems, and analyzer shelters.
- ▶ Supervised technicians responsible for preventive, corrective, and shutdown maintenance of analytical instruments.
- ▶ Led vendor evaluation and technical proposals for analyzer upgrades and modernization projects.
- ▶ Authored technical reports and engineering studies supporting analyzer reliability and procurement decisions.

Instrumentation & Control / Process Analytical Instruments Technician

2007 — 2020

Ras Lanuf Oil & Gas Processing Company (RASCO), Ras Lanuf, Libya

- ▶ Installed, calibrated, and maintained field instrumentation including transmitters, control valves, switches, and pneumatic systems across process units.
- ▶ Built hands-on expertise in HART communication, signal converters, and analyzer sample-conditioning systems.
- ▶ Provided calibration and commissioning support across multiple plant shutdowns and turnarounds.
- ▶ Gained early exposure to process analyzers, building the foundation for later analytical-instrumentation specialization.

MAJOR ENGINEERING PROJECTS

- ▶ Bently Nevada System 1 EVO Upgrade — machinery protection & vibration-monitoring modernization.
- ▶ Rosemount 5408 Radar Tank Modernization — legacy tank gauging replaced with radar-based level measurement.
- ▶ Yokogawa GC8000 Upgrade — reliability improvement of process gas chromatograph analytical trains.
- ▶ Foxboro Pneumatic Modernization — phased migration from legacy pneumatic to electronic transmitters.
- ▶ Yokogawa FLXA202 Migration — consolidation of pH and conductivity analyzers onto a unified platform.
- ▶ Aromatic Benzene Upgrade — instrumentation support for benzene measurement and process safety.
- ▶ WOMA Pump Technical Evaluation — technical evaluation supporting high-pressure pump procurement.
- ▶ Compressor Overhead Crane Restoration — engineering support restoring critical maintenance-access infrastructure.

TECHNICAL EXPERTISE

- ▶ Instrumentation & Control Systems
- ▶ Gas Chromatographs
- ▶ Shutdown Maintenance
- ▶ Process Analytical Instruments
- ▶ Preventive & Corrective Maintenance
- ▶ Commissioning & Calibration

- ▶ Control Valves, Transmitters & Switches
- ▶ Pneumatic Systems
- ▶ Vendor Communication & Procurement Support
- ▶ Reliability Improvement
- ▶ HART Communication & Signal Converters
- ▶ Technical Reports & Engineering Studies
- ▶ Technical Evaluation & Modernization
- ▶ Project Planning & Technical Leadership

ANALYTICAL INSTRUMENTS & SYSTEMS

- ▶ Siemens MAXUM Edition II (Process GC)
- ▶ Yokogawa pH Analyzer
- ▶ Ametek WDG-IV (Oxygen Analyzer)
- ▶ Sample Conditioning Systems
- ▶ Calibration Gas Systems
- ▶ Yokogawa GC8000 (Process GC)
- ▶ Yokogawa Conductivity Analyzer
- ▶ Teledyne 3020PB (H₂S Analyzer)
- ▶ Analyzer Shelters

EDUCATION & TRAINING

Diploma — Instrumentation & Control

2006

Petroleum Training & Qualifying Institute (PTQI), Tripoli, Libya

Professional Courses

- ▶ Distributed Control System (DCS) Operation & Maintenance — Arab Academy, Egypt
- ▶ Microsoft Outlook
- ▶ Pneumatic Converters
- ▶ Maintenance Management System — Train the Trainer
- ▶ Pneumatic Flow Transmitters
- ▶ Instrumentation Symbols & P&ID
- ▶ Flow & Temperature Measurement Fundamentals
- ▶ Instrumentation Electrical & Mechanical Installation
- ▶ Industrial Safety Fundamentals
- ▶ Risk Assessment
- ▶ Process Analytical Instrumentation
- ▶ Process Control Techniques
- ▶ Pneumatic Level Transmitters
- ▶ Pneumatic Pressure Transmitters
- ▶ Instrumentation Level Measurement
- ▶ Electrical & Electronic Circuits
- ▶ Pressure Measurement Fundamentals

LANGUAGES, DIGITAL SKILLS & PERSONAL INFORMATION

LANGUAGE SKILLS

Arabic: **Native**

English — Writing: **Excellent**

English — Reading: **Excellent**

English — Listening: **Very Good**

English — Speaking: **Very Good**

Certifications: **PET, FCE, IELTS**

PERSONAL INFORMATION

Date of Birth: **26 May 1988**

Nationality: **Libyan**

DIGITAL SKILLS

- ▶ Microsoft Office & Outlook
- ▶ Technical Documentation & Engineering Reports
- ▶ Website Development & CMS
- ▶ Graphic Design
- ▶ Artificial Intelligence Tools

RESIDENCE & WORK LOCATION

Residence: **Soug Al-Sabt, Libya**

Current Work Location: **Ras Lanuf City, Libya**